INCH-POUND
FF-S-85C
AMENDMENT 2
August 02, 1994
SUPERSEDING
Amendment 1
February 27, 1973

FEDERAL SPECIFICATION

SCREW, CAP, SLOTTED AND HEXAGON HEAD

This amendment forms a part of Federal Specification FF-S-85C, dated May 5, 1969, and is approved for use by all Departments and Agencies of the Department of Defense.

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* 2.1: Delete "QQ-B-637" and title.

* 2.1: Delete "QQ-B-728" and title.

* 2.1: Delete "QQ-B-750" and title.

* 2.1: Delete "QQ-C-591" and title.

* 2.1: Delete "QQ-Z-325" and title.

* 2.1: Add the following Federal Standard:
"FED-STD-H28/2 - Unified Inch Screw Threads -UN and UNR Thread Forms."

* 2.1: Delete the title for Fed. Std. No. 123 and substitute:
"Marking for Shipment (Civil Agencies)."

* 2.1: Delete "Fed. Test Method Std. No. 151" and title.

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2.1: Delete "MIL-H-3982" and title.

* 2.1: Delete "MIL-I-6866" and title.

* 2.1: Delete "MIL-I-6868" and title.

* 2.1: Delete "MIL-I-17214" and title.

* 2.1: Add the following Military Specification:
"MIL-F-18240 - Fastener Element, Self-Locking, Threaded Fastener, 250 Deg. F Maximum."

AMSC N/A FSC 5305
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* 2.1: Delete "MIL-B-24059" and title.

* 2.1: Add the following Military Standard:
"MIL-STD-1312-6 - Fastener Test Methods - Method 6, Hardness."
"MIL-STD-6866 - Inspection, Liquid Penetrant."

* 2.1: Delete "MS18153" and title.

* 2.1: Delete "MS18154" and title.

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* 2.1: Delete "MS90727" and title.

* 2.1: Delete "MS90728" and title.

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* 2.2: Delete "National Bureau of Standards (NBS) Standard:
H28, Part I - Screw-Thread Standards for Federal Services.
(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)"

2.2: Delete "United States of America Standards Institute Standards:" and substitute "American National Standards Institute (ANSI) Standards."

2.2: Delete "USAS" and substitute "ANSI" (two places).

* 2.2: Delete "USAS B18.6.2" and title and substitute "ANSI B18.6.2 - Slotted Head Cap Screws, Square Head Set Screws, and Slotted Headless Set Screws."

2.2: Delete "United States of America Standards Institute, 10 East 40th Street, New York, New York 10016" and substitute "American National Standards Institute, 1430 Broadway, New York, New York 10018".

* 2.2: Add the following ASTM standards:
* "ASTM A342 - Test Method for Permeability of Feebly Magnetic Materials."
* "ASTM B21 - Specification for Naval Brass Rod, Bar, and Shapes."
* "ASTM B98 - Specification for Copper-Silicon Alloy Rod, Bar, and Shapes."
* "ASTM B124 - Specification for Copper and Copper Alloy."
* "ASTM B138 - Specification for Manganese Bronze Rod, Bar, and Shapes."
* "ASTM B139 - Specification for Phosphor Bronze Rod, Bar, and Shapes."
* "ASTM B150 - Specification for Aluminum-Bronze Rod, Bar, and Shapes."
"ASTM B154 - Mercurous Nitrate Test for Copper and Copper Alloys."
* "ASTM B633 - Specification for Electrodeposited Coatings of Zinc on Iron and Steel."
* "ASTM D3951 - Commercial Packing, Standard Practice for."
* "ASTM E1282 - Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals and Alloys."
* "ASTM E1444 - Standard Practice for Magnetic Particle Examination."

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* "ASTM F606 - Test Methods for Determining the Mechanical Properties of Externally and Internally Threaded Fasteners, Washers and Rivets."

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* Table I: Under "Rockwell" for Grade 8, delete "C32" and "C38" and substitute "C33" and "C39", respectively.

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* 3.1.3.1: Delete "MS18153" and "MS18154" and "MS90727" and "MS90728" and substitute "ANSI B18.2.1".

Add the following new paragraphs:

3.1.5.1 Stress relief. Cold worked naval brass screws shall be stress relieved by heating the screws to a temperature of 600°F, and holding at that temperature for a minimum of 1 hour.

3.1.6 Self-locking element. Non-metallic, self-locking elements shall be in accordance with MIL-F-18240.

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* Table IA: Delete "QQ-B-728" and substitute "ASTM B138".
* Table IA: Delete "MIL-B-24059" and substitute "ASTM B150".
* Table IA: Delete "QQ-B-750" and substitute "ASTM B139".
* Table IA: Delete "QQ-C-591" and substitute "ASTM B124".
* Table IA: Delete "QQ-B-637" and substitute "ASTM B21".

Table IA: For Naval Brass, under Composition, add "alloy 462" and delete "Compostion I" and under Condition add "hard" beside alloy 462.

Table IA: For Nickel-Copper Alloy, under Condition, delete "hot finished" and under Elongation delete "30" and substitute "20".

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3.3: Delete "USAS" and substitute "ANSI".

* 3.3.1: Delete "5/8" and substitute "1/2" (two places). Delete "1-1/2" and substitute "1" (two places).

* 3.3.2: Delete "National Bureau of Standards Handbook H28, Part I" and add FED-STD-H28/2".

* 3.4.3: Delete in its entirety and substitute the following:
"3.4.3 Cadmium plating. When specified (see 6.2), cap screws shall be cadmium plated in accordance with QQ-P-416, Type II, Class 2.

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* 3.4.4: Delete "QQ-Z-325" and substitute "ASTM B633".

3.4.7: Delete in its entirety and substitute the following:
"3.4.7 Black chemical. When specified (see 6.2), brass cap screws shall be treated with a black chemical finish in accordance with MIL-F-495.

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* 3.8: Delete the first two sentences and substitute the following:
"Steel cap screws shall be permanently marked with the grade identification symbol conforming to ANSI B18.2.1, SAE J429, and applicable documents. In addition, all cap screws shall be permanently marked with the manufacturer's or private label insignia as applicable."

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Table III: Add the following to Table III - Classification of Defects:
"105 Self-locking element missing, when applicable (3.1.6) Visual".

* 4.2.4: Delete "Fed. Test Method Std. No. 151, method 111.2" and substitute "ASTM E1282".

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* 4.2.5: Delete "MIL-H-3982" and substitute "ASTM D3951".

Add the following new paragraph:
"4.2.6 Sampling for mercurous nitrate test. When specified (see 6.2), a random sample shall be taken from each lot of cold-worked naval brass screws in accordance with MIL-STD-105, Inspection Level S-1. Any one defect shall cause rejection of the represented lot."

* 4.3.2: Delete "MIL-H-3982" and substitute "ASTM D3951".

* 4.4.1: Delete "4.4.9" and substitute "4.4.13".

4.4.3: Add the following sentence: "Unless otherwise specified (see 6.2), Grade 2, low carbon steel screws shall not be subjected to this test."

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* Add the following new paragraph:
"4.4.3.4 Screws too short to be tensile strength tested shall be hardness tested in accordance with MIL-STD-1312-6. Screws shall meet the minimum hardness requirement of the material from which they were made."

* 4.4.8: Delete "MIL-I-17214" and substitute "ASTM A342".

* 4.4.9: Delete "MIL-I-6868" and substitute "ASTM E1444" and delete "MIL-I-6866" and substitute "MIL-STD-6866".

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Add the following new paragraphs:

"4.4.10 Self-locking. Self-locking screws having non-metallic elements shall be tested in accordance with MIL-F-18240."

"4.4.11 Mercurous nitrate (stress relief) test. The contractor shall furnish the Government certification that cold worked naval brass screws have been subjected to the stress relief treatment specified in 3.1.5.1. When specified (see 6.2), samples taken as specified in 4.2.6 shall be subjected to a mercurous nitrate test conducted in accordance with ASTM B154. There shall be no evidence of cracks."

* "4.4.12 Proof load. Proof load test for cap screws shall be conducted in accordance with ASTM F606."

* "4.4.13 Yield strength. Yield strength test for cap screws shall be conducted in accordance with ASTM F606."

* 5.1.1: Delete "MIL-H-3982" and substitute "ASTM D3951" (two places).

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6.2: Add the following:
"(p) Mercurous nitrate test, if required (4.4.11)."
* "(q) Proof load test, if required (4.4.12)."
* "(r) Yield strength test, if required (4.4.13)."

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - AR
Navy - SH
Air Force - 99

Review activities:
Army - AV
Navy - AS
Air Force - 82
NASA - NA
GSA - NSA

Preparing activity:
DLA - IS

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